

Hospital Management Data Analysis

Your Name

1. Introduction

This project analyzes hospital operations and billing data to identify patterns in appointments, doctor workload, treatment costs and payments. The analysis combines MySQL for data validation and business analysis with Excel PivotTables and a dashboard to present the results to a business audience.

Objective

To turn raw hospital data into clear operational and financial insights that could support decision-making.

2. Dataset

The dataset is a public hospital management dataset with five related tables covering the year 2023.

Table	Records	Contents
Patients	50	Patient and insurance details
Doctors	10	Specialization, experience, branch
Appointments	200	Patient-doctor visits and status
Treatments	200	Treatment type and cost
Billing	200	Amount, payment method and status

Patients link to appointments through `patient_id`, doctors link to appointments through `doctor_id`, treatments link to appointments through `appointment_id`, and billing links to treatments through `treatment_id`.

3. Methodology

Step 1: Data quality checks (MySQL)

Each table was checked for NULL values and duplicate IDs. Treatment costs and billing amounts were checked for negative values, and LEFT JOINS were used to confirm there were no unmatched foreign keys between the tables.

Step 2: Exploratory and business analysis (MySQL)

Queries used JOINS, GROUP BY, HAVING, aggregate functions, subqueries and date functions. The business questions covered:

- Which doctors handle the most appointments
- How appointments are distributed by status
- Which treatments generate the highest billing and have the highest average cost
- Which payment methods account for the most billing
- How billing and appointments change month by month

Step 3: Excel PivotTables and dashboard

The analysis was summarized in Excel PivotTables and PivotCharts, then combined with KPI cards and slicers (month, payment method, treatment type) into a dashboard.

4. Findings

Overall, the hospital recorded 200 appointments and a total billed amount of 551,249.85 across 50 patients and 10 doctors.

4.1 Doctor workload

Doctor D005 (Sarah Taylor) handled the most appointments with 29, followed by D001 with 25, D006 with 24, D003 with 22 and D002 with 21.

4.2 Appointment status

Status	Appointments	Share
No-show	52	26.0%
Scheduled	51	25.5%
Cancelled	51	25.5%
Completed	46	23.0%

No-show was the largest single category, but the four statuses are close in size. Only 23% of appointments were completed.

4.3 Billing and cost by treatment

Treatment	Total billed	Average cost
Chemotherapy	128,855.68	2,629.71
MRI	116,098.16	3,224.95
X-Ray	110,653.67	2,698.87
Physiotherapy	99,418.10	2,761.61
ECG	96,224.24	2,532.22

Chemotherapy generated the highest total billing, while MRI had the highest average cost per treatment.

4.4 Payment methods and status

Credit Card accounted for the highest billed amount (201,382.43), followed by Insurance (182,160.28) and Cash (167,707.14). Only 64 of the 200 bills are marked Paid, with 69 Pending and 67 Failed, so the total billed amount should not be treated as revenue received.

4.5 Monthly trend

Billing peaked in April (64,271.54) and was lowest in December (27,569.71). April also had the most appointments of any month (25).

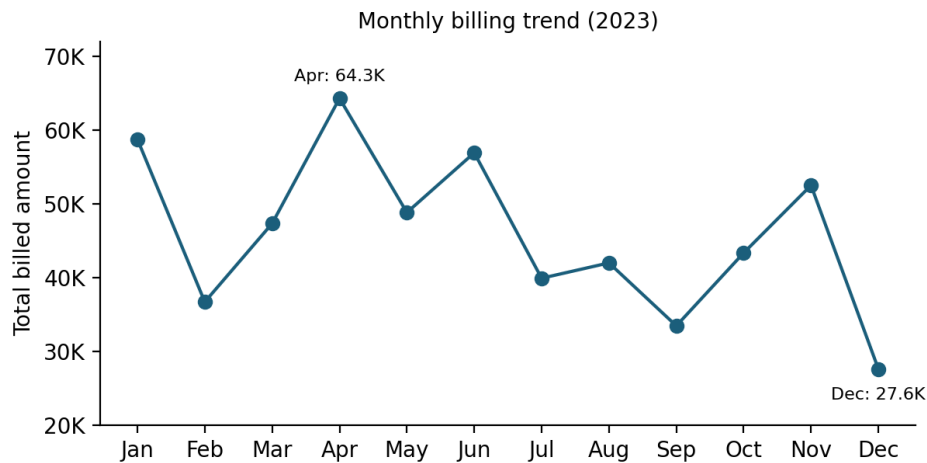


Figure 1: Monthly billing trend

4.6 Dashboard

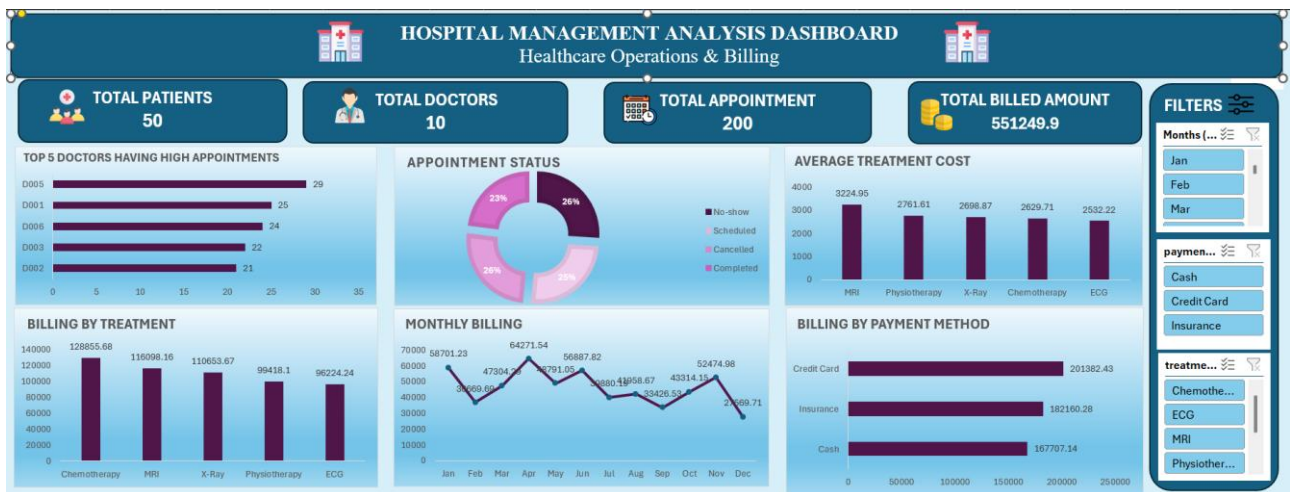


Figure 2: Excel dashboard

5. Conclusion

The project shows a complete workflow from validated relational data to a business dashboard. Key results are the high no-show share of appointments, Chemotherapy as the largest billing category, Credit Card as the top payment method, and a large share of bills that are still pending or failed. Possible next steps are appointment completion rates, payment-status analysis and a Power BI version of the dashboard.